

# 1. INTRODUCTION

## 1.1 Project Overview

The project “**Script-Controlled ACL – Restrict Record Access Based on Field Value**” is developed using the ServiceNow platform to provide secure and controlled access to records. The system uses Access Control Lists (ACLs) with server-side scripts to determine whether a user can access or modify a record.

The access decision is based on the combination of the **user's role and the value of specific record fields**. The project controls four major operations: **READ, CREATE, WRITE and DELETE**.

## 1.2 Purpose

The main purpose of this project is to prevent unauthorized access to sensitive ServiceNow records. It provides dynamic access control instead of depending only on basic role-based permissions.

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# 2. IDEATION PHASE

## 2.1 Problem Statement

In ServiceNow, controlling access only through user roles may not be sufficient when access depends on specific values in a record. Different users with the same role may require different levels of access to records.

Therefore, a mechanism is required to dynamically restrict record access based on both the **user's role and the record field value**.

## 2.2 Empathy Map Canvas

The main users of the system are ServiceNow administrators and IT support teams who manage sensitive records.

Users need secure access to records while preventing unauthorized users from viewing or modifying restricted information. They expect access control to be reliable, consistent and easy to manage.

The solution addresses their concerns by providing script-controlled ACLs that dynamically evaluate user roles and record values.

## 2.3 Brainstorming

The following ideas were considered:

- Role-based access control
- Standard ServiceNow ACLs
- Field-based access restrictions
- Script-controlled ACLs
- Separate access rules for READ, CREATE, WRITE and DELETE
- Dynamic access evaluation using JavaScript

The final solution uses **script-controlled ACLs** because they provide flexible access decisions based on both roles and field values.

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## 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey Map

The user logs into ServiceNow and attempts to access a record. The system identifies the user's role and evaluates the required ACL. The ACL script checks the relevant record field value and determines whether the requested operation is permitted.

If the conditions are satisfied, access is granted. Otherwise, the requested operation is denied.

### 3.2 Solution Requirement

#### Functional Requirements

1. **User and Role Management** – Create and manage users and roles required for access control.
2. **READ Access Control** – Allow or deny record viewing based on role and field value.
3. **CREATE Access Control** – Control whether users can create records.
4. **WRITE Access Control** – Control whether users can modify existing records.
5. **DELETE Access Control** – Control whether users can delete records.
6. **Script-Based Evaluation** – Use ServiceNow server-side JavaScript to evaluate access conditions dynamically.

#### Non-Functional Requirements

- **Security:** Prevent unauthorized record access.
- **Reliability:** Provide consistent access decisions.
- **Performance:** Execute ACL scripts efficiently.
- **Scalability:** Support additional users, roles and records.
- **Usability:** Provide simple and manageable access rules.

- **Availability:** Work within the ServiceNow cloud platform.

### 3.3 Data Flow Diagram

The system follows this basic flow:

**User → ServiceNow Interface → Authentication & Role Validation → ACL Evaluation → Role + Record Field Value Check → Access Decision → READ/CREATE/WRITE/DELETE**

Authorized requests are allowed to proceed, while unauthorized requests are rejected.

### 3.4 Technology Stack

The project uses the following technologies:

- **Platform:** ServiceNow
  - **User Interface:** ServiceNow Web UI, Forms, Lists and Service Portal
  - **Application Logic:** ServiceNow ACLs
  - **Programming:** JavaScript
  - **Database:** ServiceNow Platform Database
  - **Security:** Roles and Access Control Lists
  - **Infrastructure:** ServiceNow Cloud Platform
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## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

The project focuses on organizations and IT teams that require controlled access to sensitive ServiceNow records.

The major problems are unauthorized record access, insufficient field-level conditions and difficulty in managing different access requirements.

The proposed solution is a **Script-Controlled ACL** that evaluates the user's role together with the record's field value before allowing READ, CREATE, WRITE or DELETE operations.

### 4.2 Proposed Solution

The proposed system creates users and roles and configures ACLs for different operations. Each ACL uses conditions and server-side JavaScript to determine whether the current user is authorized.

The solution provides dynamic and centralized access control for ServiceNow records.

## 4.3 Solution Architecture

The architecture consists of the following components:

**Users → ServiceNow Web Interface → User Authentication → Role Validation → ACL Engine → Script-Controlled Access Evaluation → Record Access**

The ACL engine evaluates the user's role and relevant record field value before allowing READ, CREATE, WRITE or DELETE operations.

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## 5. PROJECT PLANNING & SCHEDULING

The project was implemented through the following milestones:

1. **Creation of Users and Roles**
2. **Creation of Required Tables**
3. **Creation of READ Access Control List**
4. **Creation of CREATE Access Control List**
5. **Creation of WRITE Access Control List**
6. **Creation of DELETE Access Control List**
7. **Testing and Conclusion**

Each milestone was implemented and verified before proceeding to the next stage.

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## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Functional Testing

Functional testing was performed to verify that the ACLs correctly control access to ServiceNow records.

The following operations were tested:

- User authentication and role validation
- READ access
- CREATE access

- WRITE access
- DELETE access
- Access restriction based on record field values
- Access restriction for unauthorized users

The expected behavior is that authorized users can perform the permitted operation, while unauthorized users are prevented from performing restricted operations.

## 6.2 Performance Testing

Performance was considered during ACL implementation by using simple server-side scripts and avoiding unnecessary database operations.

The ACL rules were designed to evaluate access conditions efficiently without introducing significant delay during normal record operations.

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## 7. RESULTS

The project successfully demonstrates script-controlled access management in ServiceNow.

The implemented ACLs provide control over:

- **READ** – Controls viewing of records.
- **CREATE** – Controls creation of records.
- **WRITE** – Controls modification of records.
- **DELETE** – Controls deletion of records.

The system evaluates the user's role and relevant record field value before allowing the requested operation. This provides more flexible and secure record-level access control.

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## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Provides dynamic access control.
- Improves protection of sensitive records.
- Supports READ, CREATE, WRITE and DELETE operations.
- Combines role-based and field-based conditions.
- Centralizes access control within ServiceNow.
- Reduces the possibility of unauthorized record modification.

## Disadvantages

- Requires knowledge of ServiceNow ACL configuration.
  - Incorrect scripts may result in unintended access restrictions.
  - Complex access requirements may require additional ACL rules.
  - Changes to access requirements may require script maintenance.
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## 9. CONCLUSION

The **Script-Controlled ACL – Restrict Record Access Based on Field Value** project provides a secure and flexible method for managing record access in ServiceNow.

By combining user roles with record field values, the system can dynamically determine whether a user is allowed to READ, CREATE, WRITE or DELETE a record. The implementation demonstrates how ServiceNow ACLs and server-side JavaScript can be used to strengthen record-level security and provide controlled access to sensitive information.

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## 10. FUTURE SCOPE

The project can be enhanced in the future by:

- Adding more complex field-based access conditions.
  - Supporting department-based and location-based access.
  - Creating reusable ACL scripts for multiple tables.
  - Adding detailed access-denial logging.
  - Extending the solution to additional ServiceNow applications.
  - Integrating advanced security and compliance monitoring.
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## 11. APPENDIX

### Source Code

The project contains ServiceNow ACL configurations and server-side JavaScript used for access evaluation.

### Project Components

- User and Role Configuration
- ServiceNow Tables
- READ ACL
- CREATE ACL
- WRITE ACL
- DELETE ACL

## Project Demo

The completed project demonstrates controlled access to ServiceNow records based on user roles and record field values.

**GitHub link:** <https://github.com/aswini-sivakumar/Vetri-Thiran-payirchi-Thittam>

**Project Demo link:** [https://drive.google.com/file/d/1h-2ila0W-QP3dDFzo8TQ9XYX\\_MxdWZIZ/view?usp=drivesdk](https://drive.google.com/file/d/1h-2ila0W-QP3dDFzo8TQ9XYX_MxdWZIZ/view?usp=drivesdk)